Experiment 7 : Time Series Analysis of Stock Market Data

Aim: To perform time series analysis on stock price data to identify trends, patterns and seasona to build a simple model to forecast future stock prices.

In [1]:
```
# A52 Dipanshu Ambilkar
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
np.random.seed(6)
dates = pd.date_range("2025-01-01", periods=240)
price = 100 + np.cumsum(np.random.normal(0.3, 2, 240))
df = pd.DataFrame({
    "date": dates,
    "close": np.round(price, 2),
    "volume": np.random.randint(100000, 900000, 240),
})
df.head()
```

Out[2]:

	date	close	volume
0	2025-01-01	99.68	592896
1	2025-01-02	101.43	231931
2	2025-01-03	102.17	751394
3	2025-01-04	100.67	260365
4	2025-01-05	96.00	420685

In [3]:
```
print("total rows:", df.shape)
```

```
total rows: (240, 3)
```

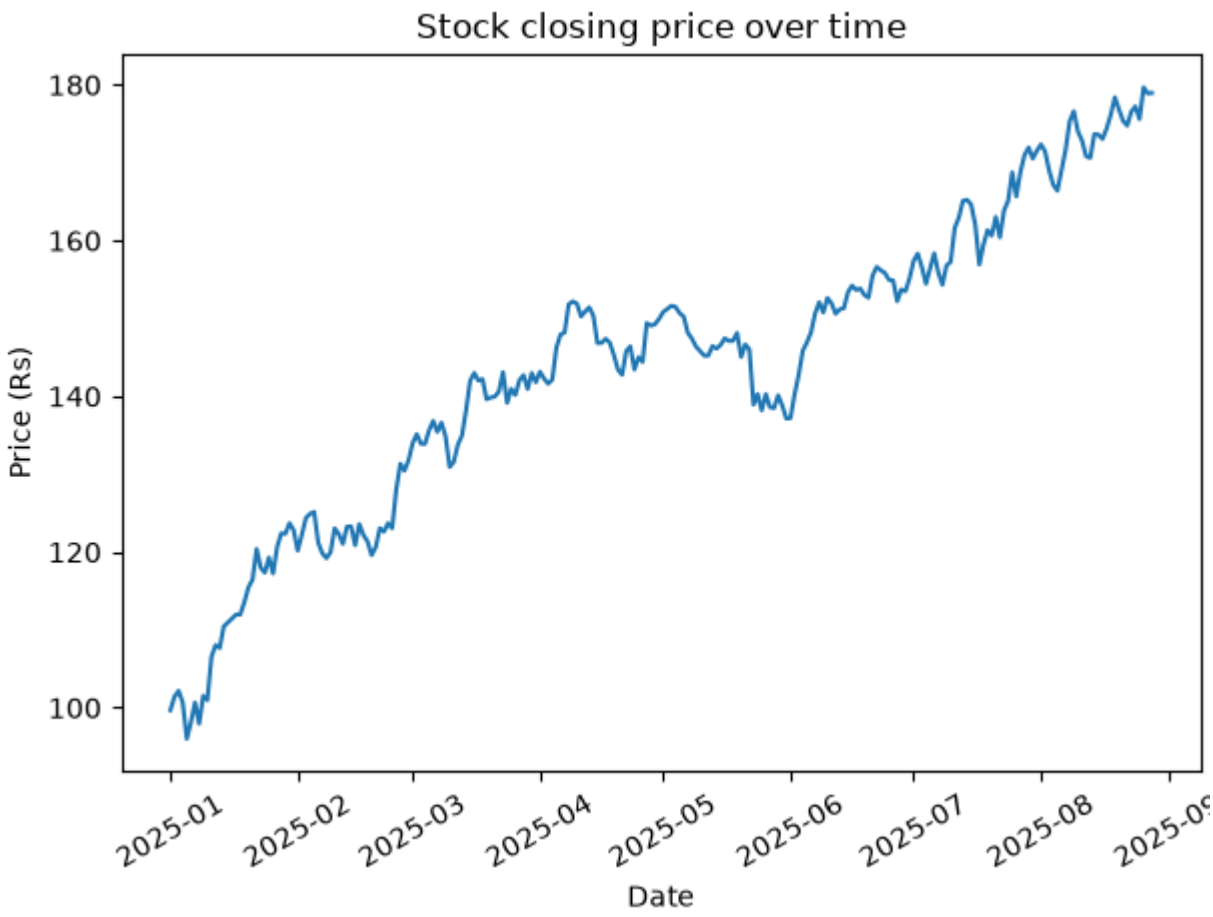
In [4]:
```
df["date"] = pd.to_datetime(df["date"])
df = df.set_index("date")
df.head()
```

Out[4]:

	close	volume
date		
2025-01-01	99.68	592896
2025-01-02	101.43	231931
2025-01-03	102.17	751394
2025-01-04	100.67	260365
2025-01-05	96.00	420685

In [5]:
```
plt.plot(df.index, df["close"])
plt.title("Stock closing price over time")
```

```
plt.xlabel("Date")
plt.ylabel("Price (Rs)")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



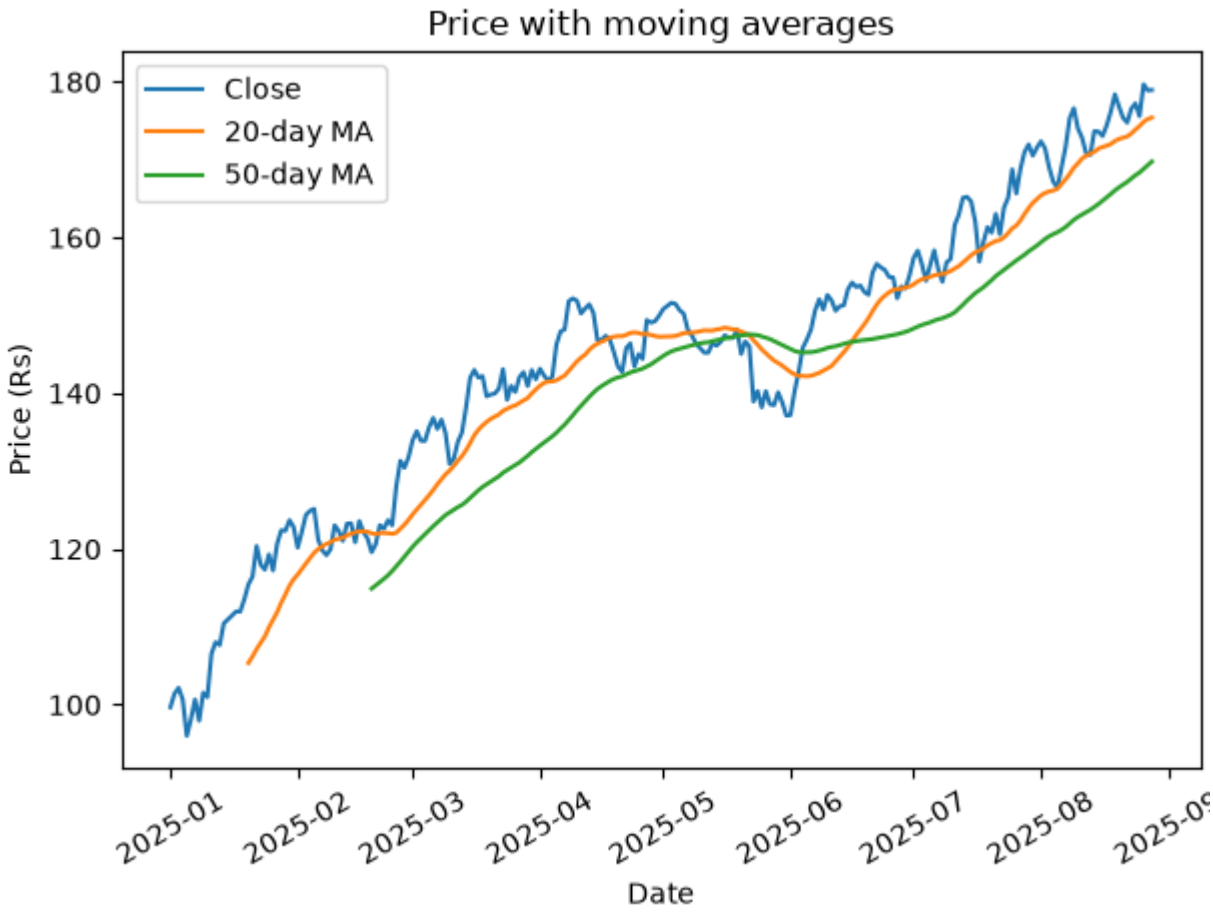
In [6]:
```
df["ma20"] = df["close"].rolling(20).mean()
df["ma50"] = df["close"].rolling(50).mean()
df.head(55).tail()
```

Out[6]:

date	close	volume	ma20	ma50
2025-02-20	120.61	285589	121.9040	115.326
2025-02-21	123.05	221667	122.0480	115.758
2025-02-22	122.60	673705	122.0690	116.167
2025-02-23	123.70	573534	122.0355	116.627
2025-02-24	123.05	178243	121.9430	117.168

In [7]:
```
plt.plot(df.index, df["close"], label="Close")
plt.plot(df.index, df["ma20"], label="20-day MA")
plt.plot(df.index, df["ma50"], label="50-day MA")
plt.title("Price with moving averages")
plt.xlabel("Date")
plt.ylabel("Price (Rs)")
plt.legend()
plt.xticks(rotation=30)
```

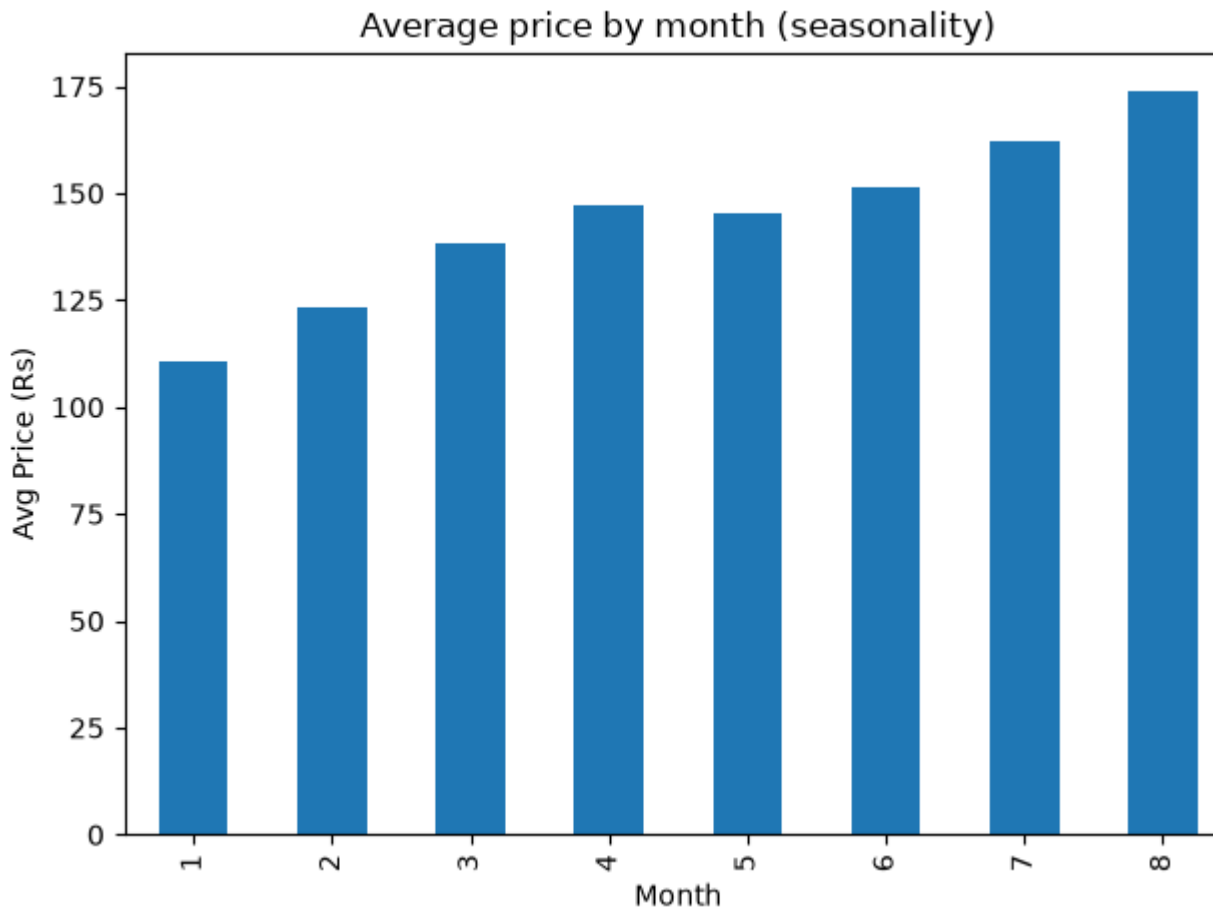
```
plt.tight_layout()
plt.show()
```



In [8]:
```
df["month"] = df.index.month
monthly = df.groupby("month")["close"].mean()
print(monthly)
```

```
month
1    110.577742
2    123.288571
3    138.346774
4    147.082333
5    145.217097
6    151.381667
7    162.137097
8    173.938929
Name: close, dtype: float64
```

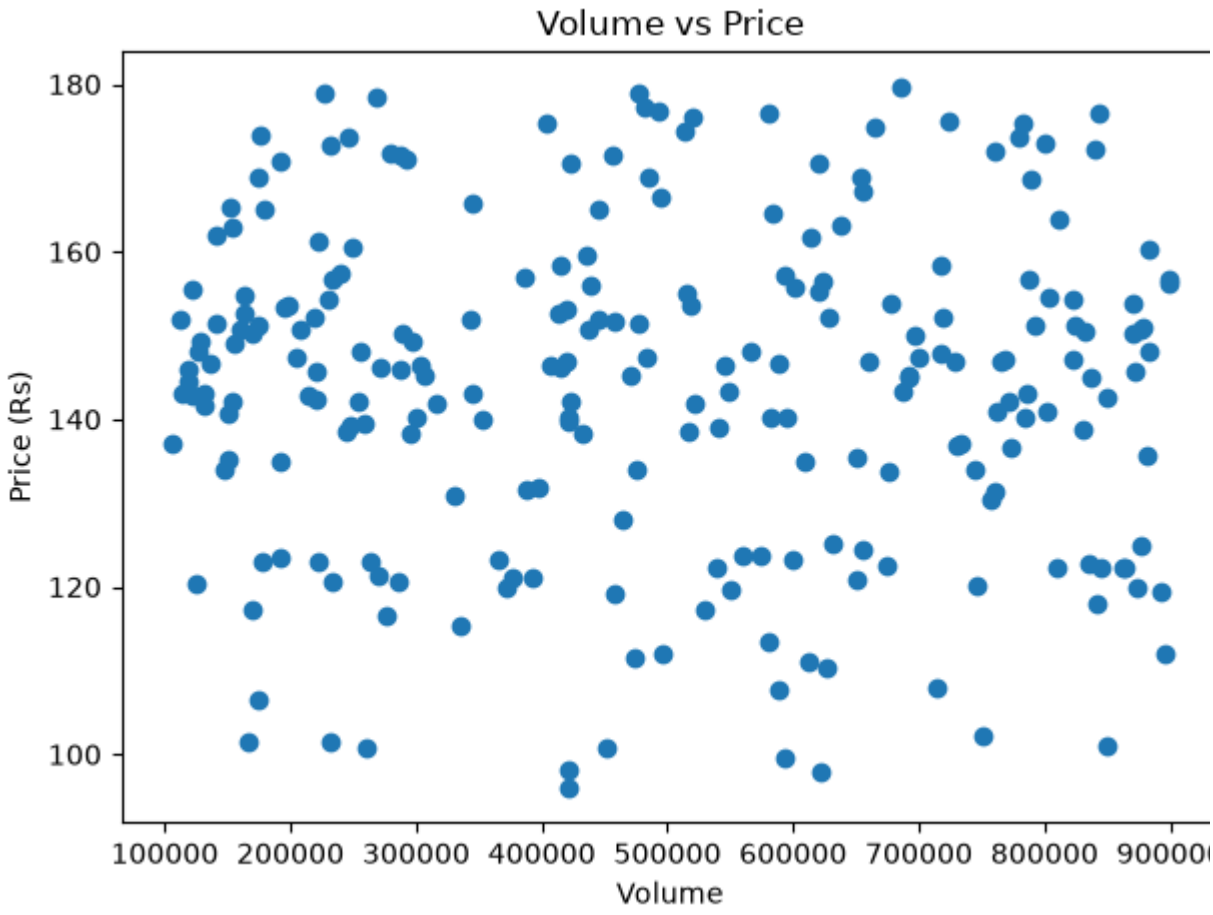
In [9]:
```
monthly.plot(kind="bar")
plt.title("Average price by month (seasonality)")
plt.xlabel("Month")
plt.ylabel("Avg Price (Rs)")
plt.tight_layout()
plt.show()
```



In [10]:
```
print("price-volume correlation:", df["close"].corr(df["volume"]))
```

```
price-volume correlation: -0.037906536412319884
```

In [11]:
```
plt.scatter(df["volume"], df["close"])
plt.title("Volume vs Price")
plt.xlabel("Volume")
plt.ylabel("Price (Rs)")
plt.tight_layout()
plt.show()
```



In [12]:
```
alpha = 0.3
s = [df["close"].iloc[0]]
for p in df["close"].iloc[1:]:
    s.append(alpha * p + (1 - alpha) * s[-1])
df["forecast"] = s
df.tail()
```

Out[12]:

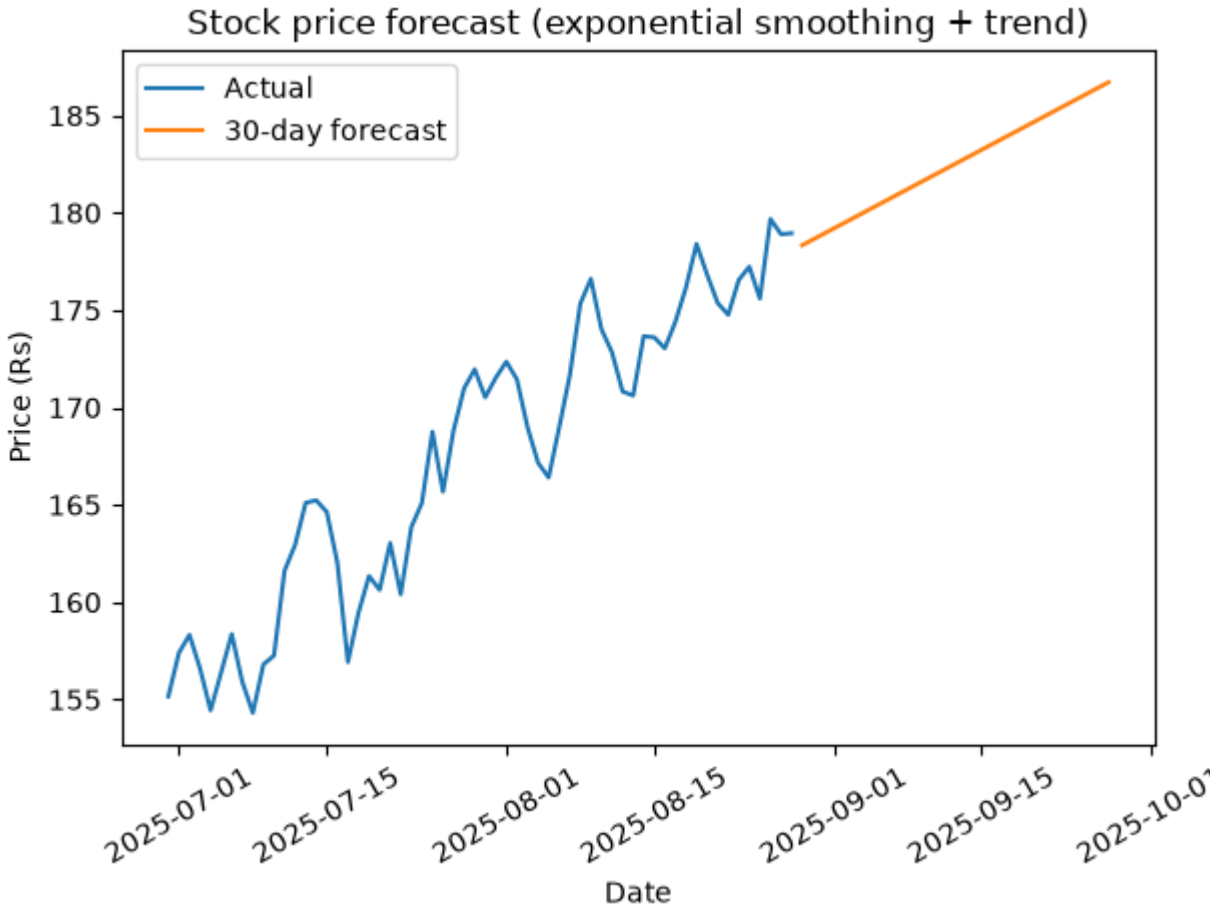
date	close	volume	ma 2 0	ma 5
2025-08-24	177.23	4823044	173.8615	168.005
2025-08-25	175.59	7241366	174.3210	168.350
2025-08-26	179.68	6857033	174.8565	168.825
2025-08-27	178.90	4776922	175.2195	169.317
2025-08-28	178.95	2275448	175.4005	169.760

In [13]:
```
future = pd.date_range(df.index[-1] + pd.Timedelta(days=1), periods=30)
last = s[-1]
trend = (s[-1] - s[-30]) / 30
fp = [last + trend * i for i in range(1, 31)]
print("next 5 forecast values:", [round(x, 2) for x in fp[:5]])
```

```
next 5 forecast values: [np.float64(178.33), np.float64(178.62),
np.float64(178.91), np.float64(179.2), np.float64(179.49)]
```

In [14]:
```
plt.plot(df.index[-60:], df["close"][-60:], label="Actual")
plt.plot(future, fp, label="30-day forecast")
```

```
plt.title("Stock price forecast (exponential smoothing + trend)")
plt.xlabel("Date")
plt.ylabel("Price (Rs)")
plt.legend()
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



In [15]:
```
print("done")
```

```
done
```